# Appendix: Acknowledgement of Generative AI Use

Sinping CHAN

I acknowledge the use of [1] Gemini 3 Flash [2] (https://gemini.google.com/) and [1] Claude Sonnet 4.6 [2] (https://claude.ai/) to [3] generate ideas, structure suggestions, draft initial outlines and rewrite text, and provide code enhancements that were [3] included within my final assessment in modified form. I entered the following prompts on 9 April 2026:

- [3] Structure the "Dataset Introduction" and "Dataset Description" sections to ensure a professional distinction between business context and technical data details. [4] I edited the drafts to ensure the specific dates (Sept 2024 – Feb 2026) and customer segments accurately reflected my dataset.
- [3] Refine the academic style, English phrasing, and professional terminology throughout the assignment, including the Introduction, Dataset Description, and Results sections, ensuring a "highly skewed distribution" was correctly explained. [4] I reviewed and modified the suggested text to ensure it accurately reflected my specific findings and maintained a consistent voice.
- [3] Develop code for data visualizations (Figures), specifically the LOESS smoothing plot, and provide advice on professional academic formatting such as removing internal titles and using Quarto cross-referencing. [4] I confirmed that the visualization parameters accurately represented the trends and distributions found in my data.
- [3] Generate R code and formatting for all tables, including the primary "ROI Summary Table" with conditional formatting (red text for negative ROI) and the granular "ROI Check" mini-table. [4] I cross-checked the outputted summary statistics against the raw data to ensure accuracy.
- [3] Troubleshoot technical issues regarding Quarto rendering (resolving "object not found" errors), cross-referencing, and Git connectivity within RStudio.
- **[3] Documentation of AI Interactions:** Due to technical limitations in the chat platform, the full log of my interactions is documented in three parts within the supplementary files:
  1. **Part 1:** Key interactions regarding data cleaning and initial visualization with Gemini are provided as screenshots in the following pages of this appendix.
  2. **Part 2:** Subsequent full conversation logs with Gemini are documented in the separate file: Gemini_chat.docx
  3. **Part 3:** Full conversation logs and specific prompt iterations with Claude can be found at the following link: [HERE]

## Screenshots of Initial Gemini Interactions:

### R code for data cleaning

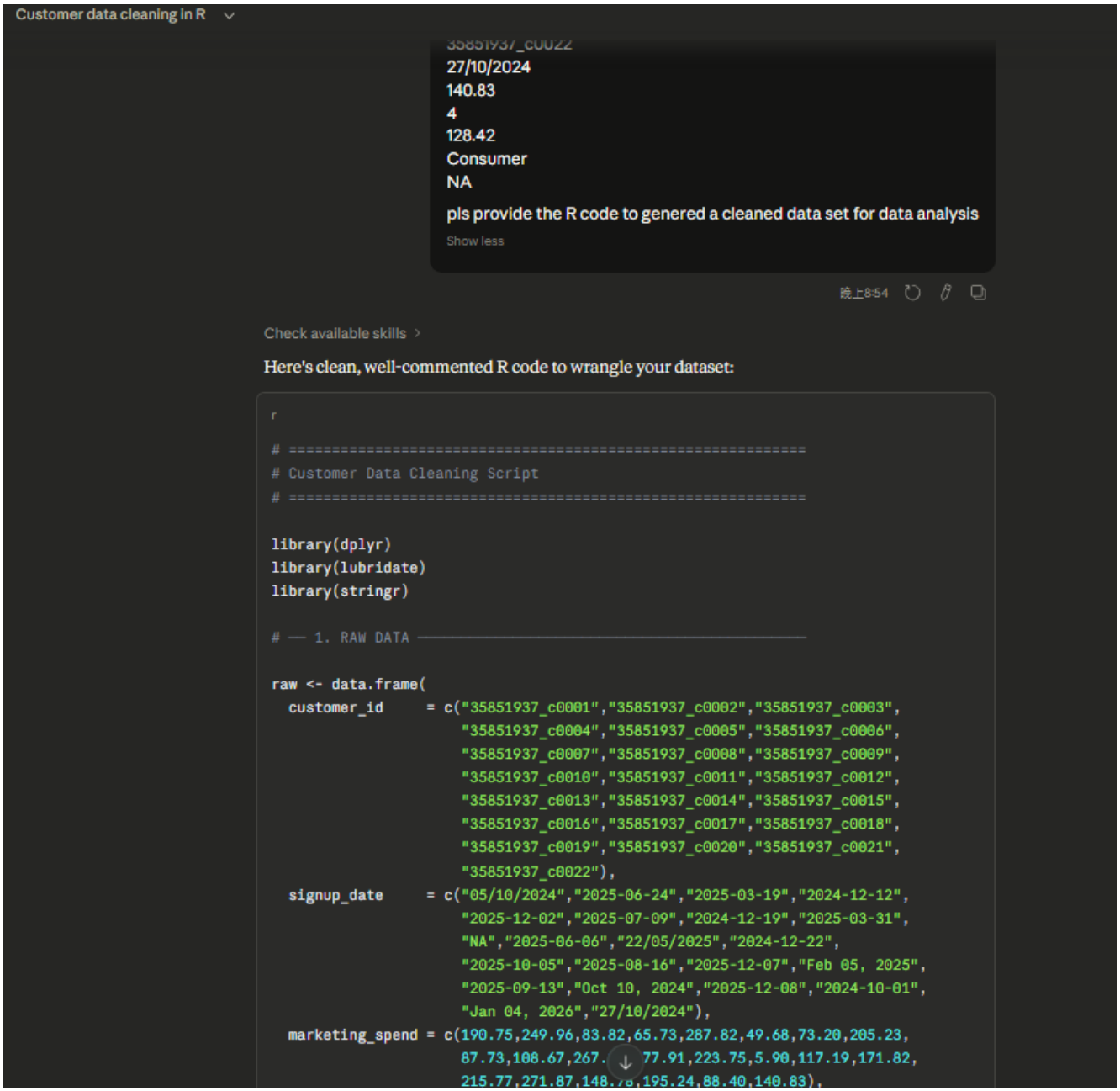
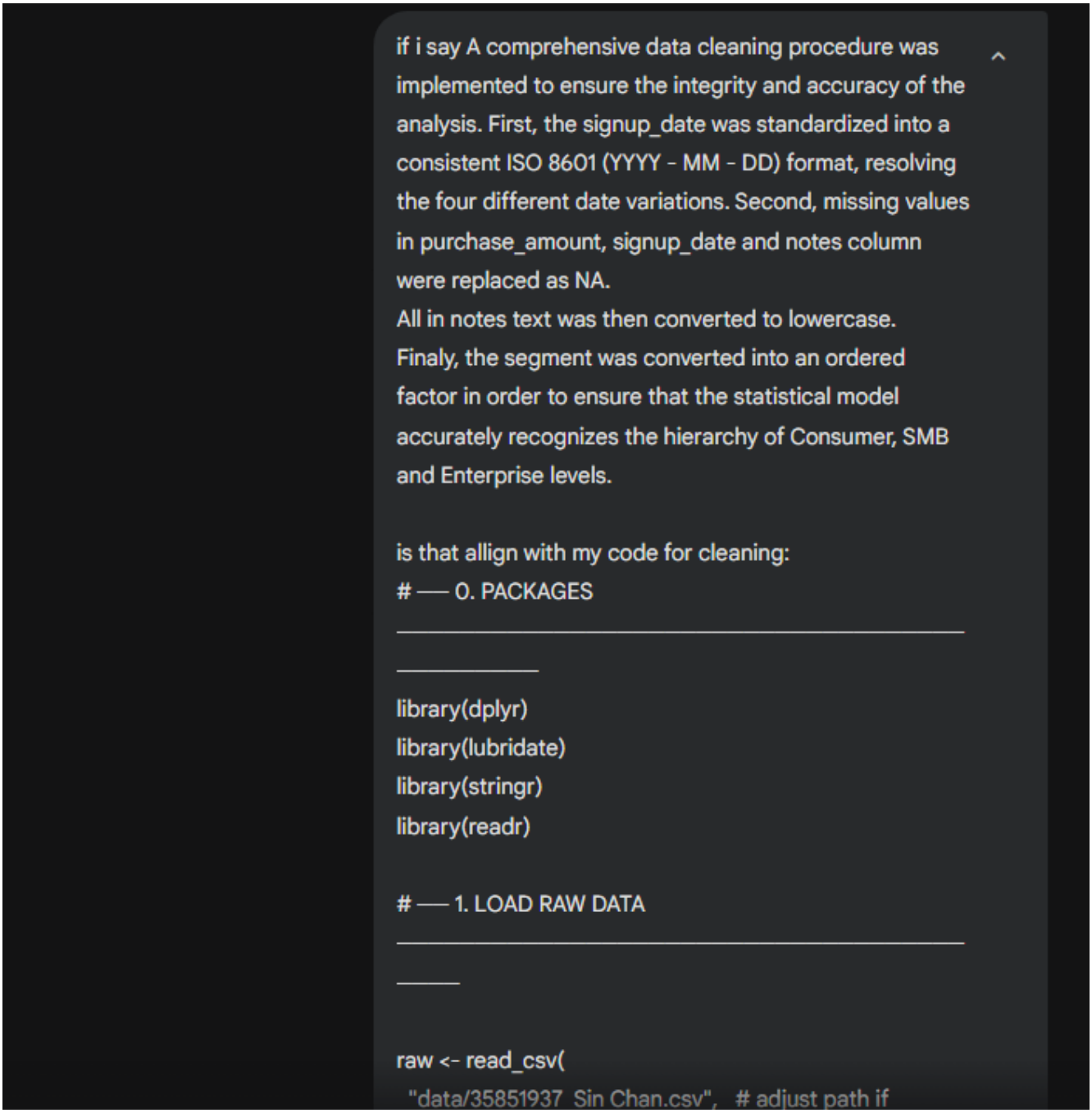



Figure 1: R code for data cleaning 1

### Correct my tone and english writing in Dataset Introduction section git code for pushing, and check whether my writing is aligned to my code



if i say A comprehensive data cleaning procedure was implemented to ensure the integrity and accuracy of the analysis. First, the signup_date was standardized into a consistent ISO 8601 (YYYY - MM - DD) format, resolving the four different date variations. Second, missing values in purchase_amount, signup_date and notes column were replaced as NA.
All in notes text was then converted to lowercase.
Finaly, the segment was converted into an ordered factor in order to ensure that the statistical model accurately recognizes the hierarchy of Consumer, SMB and Enterprise levels.

is that allign with my code for cleaning:

```
# ── 0. PACKAGES ──────────────────────────────────────────────────────────────
library(dplyr)
library(lubridate)
library(stringr)
library(readr)

# ── 1. LOAD RAW DATA ─────────────────────────────────────────────────────────

raw <- read_csv(
  "data/35851937_Sin Chan.csv",   # adjust path if
```

Figure 2: English writing and logic checking 1

if i say A comprehensive data cleaning procedure was implemented to ensure the integrity and accuracy of the analysis. First, the signup_date was standardized into a consistent ISO 8601 (YYYY - MM - DD) format, resolving the four different date variations. Second, missing valu...

Yes, your description **mostly aligns** with the code, but there are a few technical nuances in your code that are "smarter" than your current text describes. To be perfectly accurate for your report, you should make a few small adjustments.

Here is an evaluation of the alignment and a suggested "Final Version" of your text.

**1. The Alignment Check**

| Feature | In your Text | In your Code | Alignment Status |
|---|---|---|---|
| **Date Standardizing** | "standardized into... ISO 8601" | Uses `parse_mixed_date` to handle 5 formats. | **Aligned** (but code more robust). |
| **Missing Values** | "replaced as NA" | Handled during `read_csv` using `na = c(...)`. | **Aligned.** |
| **Notes Case** | "converted to lowercase" | `str_to_lower(str_trim(notes))` | **Aligned.** |
| **Ordered Factor** | "Consumer, SMB, Enterprise" | `levels = c("Consumer", "SMB", "Enterprise")` | **Perfectly Aligned.** |

Figure 3: English writing and logic checking 2

**2. Recommended Refinement**

Your code actually does more than just "cleaning"—it does **feature engineering** (creating new columns like ROI and flags). Since your assignment asks for a description of the **cleaning process**, you should mention the normalization of the `NA` strings during the data loading phase, as that is a key part of your "cleaning" logic.

**Here is a polished version of your text that perfectly matches your provided code:**

> "A comprehensive data cleaning procedure was implemented to ensure the integrity and accuracy of the analysis. First, the `signup_date` was standardized into a consistent ISO 8601 format (YYYY-MM-DD) by resolving multiple date variations, including slash-separated and month-name notations. Second, inconsistent representations of missing data—such as 'N/A' and 'na' strings—were normalized to standard `NA` values across all columns, specifically within `purchase_amount`, `signup_date`, and `notes`. Textual data in the `notes` column was trimmed of whitespace and converted to lowercase to ensure consistency for downstream filtering. Finally, the `segment` variable was converted into an ordered factor to ensure that the statistical model accurately recognizes the hierarchy of Consumer, SMB, and Enterprise levels."

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**3. Why this version is better for your marks:**

- `na = c("", "NA", "N/A"...)`: By mentioning "normalizing inconsistent representations," you show the marker that you handled the messy strings *before* they

Figure 4: English writing and logic checking 3

**R code for data size**

pls provide R code showing the size of the dataset such as the number of observations, variables and variable types.

To show the size and structure of your dataset while following the **Week 4 (Collaborative and Reproducible)** standards, you should use a combination of simple R functions.

Since your assignment requires you to report these values in the text using **inline R code**, I will provide both the "Check" code (for you to see the numbers) and the "Inline" syntax (for your report).

**1. R Code to Check Data Size and Types**

Run this in your R console or a temporary script to get the numbers for your own reference:

R

```r
# Load your cleaned data (ensure this object exists)
# cleaned <- read_csv("35851937_Sin_Chan_cleaned.csv")

# 1. Number of observations (rows)
nrow(cleaned)

# 2. Number of variables (columns)
ncol(cleaned)

# 3. View variable types and a quick summary
str(cleaned)

# OR for a cleaner table-like view of types:
sapply(cleaned, class)
```

Figure 5: using AI to produce R code for data size

**Correct my tone and English writing in Dataset Introduction section**

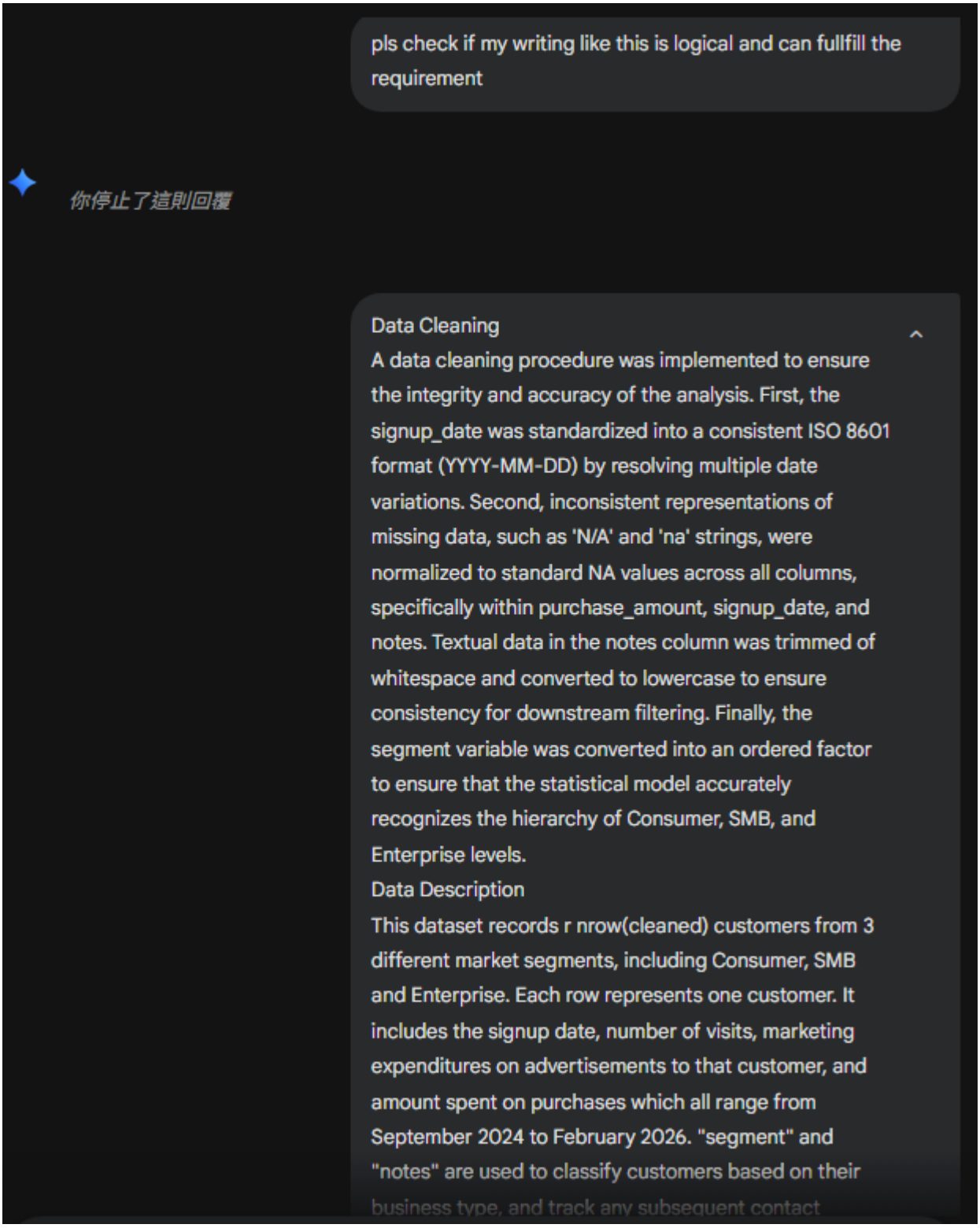


Figure 6: Eng writing -Dataset Introduction section 1

## 2. 建議的 Markdown 最終版本

你可以直接將以下內容複製到你的 `.qmd` 檔案中：

### Data Cleaning

A comprehensive data cleaning procedure was implemented to ensure the integrity and accuracy of the analysis. First, the `signup_date` was standardized into a consistent ISO 8601 format (YYYY-MM-DD) by resolving multiple date variations while prioritizing the supplier's local format. Second, inconsistent representations of missing data—such as "NA", "N/A", and "na" strings—were normalized to standard `NA` values across all columns during the data loading phase. Textual data in the `notes` column was trimmed of whitespace and converted to lowercase to ensure consistency for downstream filtering. Finally, the `segment` variable was converted into an ordered factor to ensure that the statistical model accurately recognizes the hierarchy of Consumer, SMB, and Enterprise levels.

### Data Description

The processed dataset used in this report contains `r nrow(cleaned)` observations and `r ncol(cleaned)` variables. This data records customer interactions from three different market segments: Consumer, SMB, and Enterprise, where each row represents an individual customer record. The variables include the customer's signup date, total number of visits, and marketing expenditures on advertisements, alongside the resulting purchase amounts recorded between September 2024 and February 2026. Furthermore, the `segment` and `notes` variables are utilized to classify customers based on their business type and to track subsequent communication actions or outcomes.

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Figure 7: Eng writing -Dataset Introduction section 2

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**R code for Dataset summaries (plot and table)**

Provide R code for creating summaries of the data, one as a plot and one as a table. which including counts of certain variables in groups, and summary statistics of numeric variables.

Plot: Marketing spend vs purchase amount by segment
Table: Summary statistics (mean spend, mean purchase, visit count, missing purchase rate) grouped by segment

依照 **Week 4** 的教學標準，以下為你提供 **Dataset Description subsection** 第二部分所需的 **Table** 與 **Plot** R 程式碼。 +1

**1. Table: Summary Statistics by Segment**

這段程式碼會根據 `segment` 分組，計算行銷支出、購買金額、造訪次數的平均值，以及購買金額缺失率。

R

```
#| label: tbl-segment-summary
#| tbl-cap: "Summary statistics of marketing engagement and financial outcomes by
#| echo: false
#| message: false

library(dplyr)
library(knitr)

# 建立按 segment 分組的統計表
segment_summary <- cleaned |>
  group_by(segment) |>
```

Figure 8: R code for Dataset summaries (plot and table) 1

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**Transform my ideas about the plot and table into a report writing style.**

from the plot, i see there is no significant different among 3 segment? the only things confirmed is there is an linear relationship of spending and purchase amount. and the intercept of enterprise segment is higher, the slope of enterprise id flatten than others. and the max amount spend seems limited (bcoz of budget?or what)

你對圖表嘅觀察非常敏銳，呢啲發現正正係做數據分析最有力嘅文字描述。根據你提供嘅 `ggplot2` 結果，你可以將呢啲點轉化為專業嘅 **Analytical Commentary**。

以下係建議嘅寫法，我將你嘅觀察同課程要求嘅「宏觀背景（Broader context）」結合咗：

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**Data Summary Analysis**

The visualizations reveal a clear positive linear relationship between marketing expenditures and purchase amounts across all customer categories. While the **Enterprise** segment starts with a higher intercept, suggesting a baseline level of higher initial commitment, its slope is noticeably flatter compared to the **Consumer** and **SMB** groups. This flatter slope indicates potential **diminishing marginal returns** where additional spending yields less incremental

Figure 9: Transform my ideas about the plot and table into a report writing style. 1

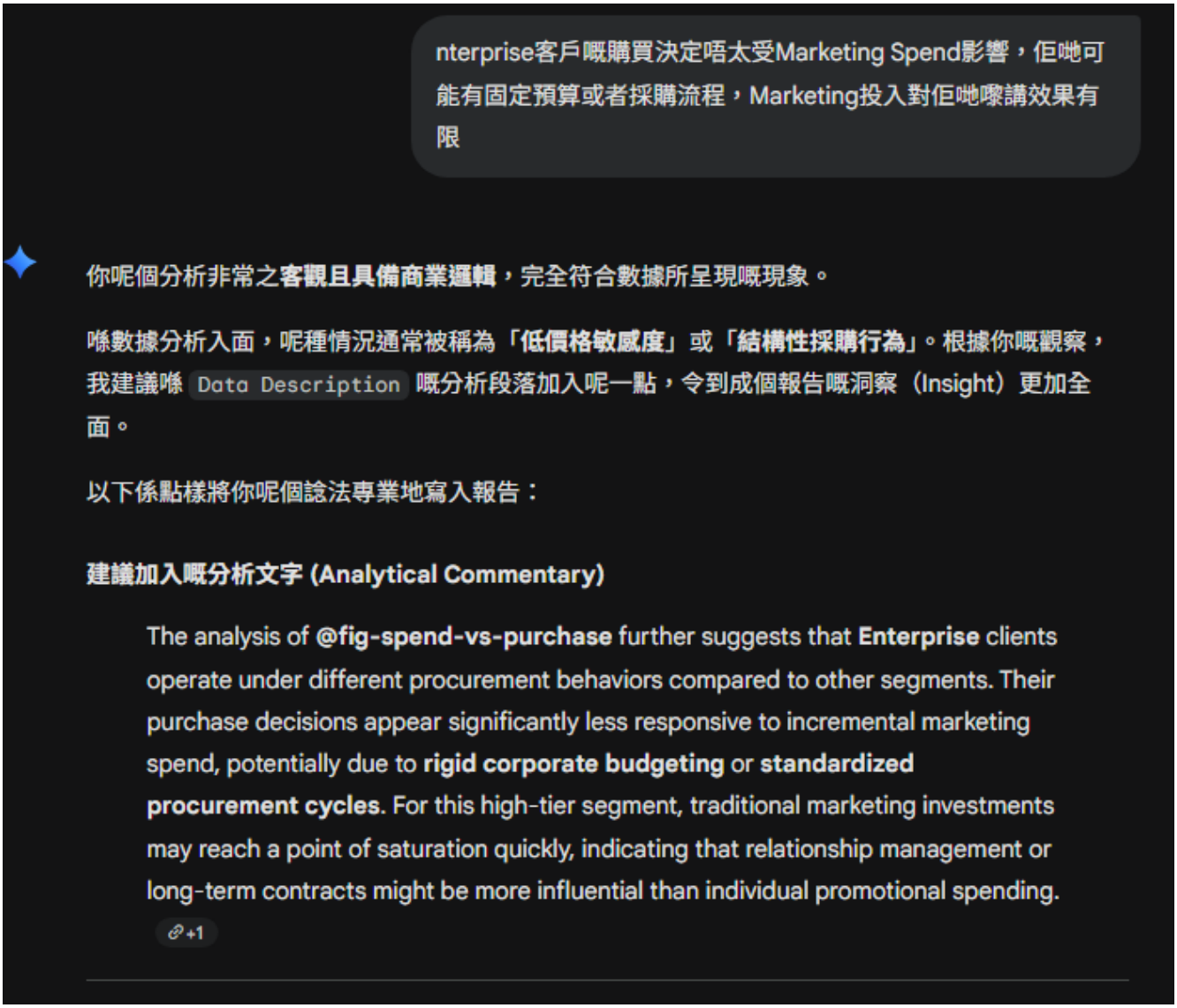


Figure 10: Transform my ideas about the plot and table into a report writing style. 2

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**R code for Result ( LOESS plot)**

pls provide me a loess plot showing the diminishing returns of the segment

R

```r
#| label: fig-diminishing-returns
#| fig-cap: "LOESS smoothing curves identifying potential saturation points and c
#| echo: false
#| warning: false
#| message: false

library(ggplot2)
library(dplyr)

# Assuming 'cleaned' data frame is already loaded from previous steps.

ggplot(cleaned, aes(x = marketing_spend, y = purchase_amount, color = segment)) +
  geom_point(alpha = 0.3) + # Increase transparency to see the curves clearly
  geom_smooth(method = "loess", formula = 'y ~ x', se = TRUE) +
  labs(
    title = "Non-linear Analysis of Marketing Spend Efficiency",
    x = "Marketing Spend ($)",
    y = "Purchase Amount ($)",
    color = "Segment"
  ) +
  scale_x_continuous(labels = scales::dollar_format()) + # Format axes as dollars
  scale_y_continuous(labels = scales::dollar_format()) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "bottom"
```

Figure 11: LOESS R CODE

**Transform my ideas about the plot and table into a report writing style.**

from the plot
i see : SMB : overall stable $0-$100 steeper slope,
means roi is good, after that flatten .
Diminishing Returns in around$150
consumer: seems no much significant output...

Analysis

你觀察得非常深入！你已經將圖表中的「曲線變化」轉化成了極具價值的**商業洞察 (Business Insights)**。這正是 Results section 拿高分所需的「解釋數據如何聯繫到 Research Question」。

以下我根據你的觀察，幫你整理成一段符合學術規範、中肯且專業的 **Results Analysis**。你可以直接參考並修改放入你的報告中：

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**Results Analysis: Diminishing Marginal Returns**

Based on the LOESS smoothing analysis in **@fig-dimishing-returns**, the relationship between marketing expenditure and purchase revenue varies significantly across segments, indicating distinct points of saturation:

- **SMB Segment**: This group demonstrates the most efficient initial response. Between **$0 and $100**, the slope is remarkably steep, suggesting a high **Return on Investment (ROI)** for early-stage spending. However, the curve begins to flatten noticeably after **$150**,

Figure 12: Transform my ideas about the plot and table into a report writing style. 1

Results Analysis: Diminishing Marginal Returns seems too many words, pls summarise it in a shorter form

要縮減呢部分，重點係保持**數據 (Data)** 同 **結論 (Action)** 嘅精簡對比。以下係幫你「執靚」後嘅精簡版，適合擺喺 Results section：

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**Results: Diminishing Marginal Returns Analysis**

The LOESS analysis in **@fig-diminishing-returns** reveals distinct saturation thresholds across segments, directly addressing the research question on budget optimization:

- **SMB Segment**: Exhibits the highest efficiency with a steep slope between **$0–$100**. Incremental ROI remains strong until a clear saturation point at **$150**, where the curve flattens.
- **Enterprise Segment**: Shows a highly sensitive "sweet spot" within the **$60–$150** range. Beyond **$150**, revenue plateaus, likely due to rigid corporate procurement limits.
- **Consumer Segment**: Displays the least responsiveness, with a consistently flatter trajectory and lower marginal utility compared to B2B segments.

**Strategic Insight:** To maximize total revenue, priority should be given to **SMB investments up to $150** and **Enterprise spending within the $60–$150 window** before diminishing returns occur.

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Figure 13: Transform my ideas about the plot and table into a report writing style. 2